Homework 09, Week 12 2023.11.22 Due 2023.11.30 J20.01, J20.06, J20.12, L09.09, L09.10

★ J20.01

Demonstrate that each term of the generating functions $U_1(q, Q)$, $U_2(q, P)$, $U_3(p, Q)$ and $U_4(p, P)$ must have the units of action.

Solution

★ J20.06

The generating function $U_2(q, P) = \sum_{k=1}^{D} g_k(q)\, P_k$ generates transformation of Lagrangian type, defined as

transformations with $Q_k = Q_k(q)$ only. Now consider a generating function $U_2(q, P) = \sum_{k=1}^{D} q_k\, f_k(P)$

(a) What condition must be imposed on the matrix with elements $\partial f_k(P)/\partial P_l$ in order for this to be a legitimate generating function?

(b) Show that in this transformation, the new momenta P_k are functions only of the old momenta p

(c) Is this transformation of the Lagrangian type? If so why, if not why not?

Solution

★ J20.12 & J20.13

Consider a system consisting of a single mass with phase space variables q, $p = x, y, z, p_x, p_y, p_z$.

Let $\hat{\boldsymbol{n}}$ be an arbitrary, constant unit vector. $\vec{\boldsymbol{L}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}}$ is the angular momentum vector.

(a) Show that the differential generating function （课件中称为无穷小正则变换生成元，微分生成元）

$G = \hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{p}}$ generating a translation $\delta \vec{\boldsymbol{r}} = \epsilon\, \hat{\boldsymbol{n}}$ with no change in $\vec{\boldsymbol{p}}$.

(b) Show that the differential generating function $G = -\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{r}}$ generates a change of momentum $\delta \vec{\boldsymbol{p}} = \epsilon\, \hat{\boldsymbol{n}}$ with no change in position.

(c) Show that the differential generating function $G = \hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{L}}$ generates changes of position and momentum

$\delta \vec{\boldsymbol{r}} = \epsilon\, \hat{\boldsymbol{n}} \times \vec{\boldsymbol{r}}, \quad \delta \vec{\boldsymbol{p}} = \epsilon\, \hat{\boldsymbol{n}} \times \vec{\boldsymbol{p}}$

(d) Show that the change given in step (c) denotes a differential rotation of $\vec{\boldsymbol{r}}$ and $\vec{\boldsymbol{p}}$ by angle ϵ about axis $\vec{\boldsymbol{n}}$

Solution

★ L09.09

已知生成函数 $U(q, P, t) = (q_1 \cos\omega t + q_2 \sin\omega t)\, P_1 + (-q_1 \sin\omega t + q_2 \cos\omega t)\, P_2 + q_3\, P_3$，求正则变换。

如果哈密顿函数 $H = \frac{1}{2m}\left(p_1^2 + p_2^2 + p_3^2\right) + V(r)$，其中 $r^2 = q_1^2 + q_2^2 + q_3^2$，求变换后的哈密顿函数 $K(Q, P, t)$

Solution

★ L09.10

设系统的哈密顿量 $H = \frac{1}{2}\left(X q^2 + 2 Y q p + Z p^2\right)$，其中$q$，$p$ 分别为系统的广义坐标与广义动量，(X, Y, Z) 是常参数且 $X Z > Y^2$。

(i) 证明系统作简谐振动，振动频率为 $\omega = \sqrt{X Z - Y^2}$

(ii) 求出作用量变量 J，并证明它可以表示为 $J = \frac{2\pi E}{\omega}$，其中 E 为系统的机械能

(iii) 做变换 $q = \sqrt{Z}\, Q$ 且变换是正则的，变换后的系统哈密顿量为：$K = \frac{1}{2} P^2 + \frac{1}{2}\omega^2 Q^2$，求出 P 与 q，p 的关系以及生成函数。

Solution